

TCP Guide for Thaghrah

Description

TCP (Transmission Control Protocol) is one of the main protocols of the Internet Protocol Suite (TCP/IP). It operates at the **Transport Layer** and provides **reliable, ordered, and error-checked** delivery of data between applications running on different hosts.

Unlike UDP, TCP is **connection-oriented** — it establishes a connection before sending data, ensures data arrives correctly and in the right order, and gracefully closes the connection when finished. It is widely used by applications that require reliability, such as web browsing (HTTP), email (SMTP), file transfer (FTP), and remote login.

TCP achieves reliability through mechanisms like **sequence numbers**, **acknowledgments**, **retransmission** of lost packets, **flow control**, and **congestion control**.

Key Components of TCP

1. TCP Connection Establishment — The 3-Way Handshake

Before any data is sent, TCP performs a **three-way handshake** to synchronize sequence numbers and establish a reliable connection:

- **Step 1:** Client sends a packet with the **SYN** (Synchronize) flag.
- **Step 2:** Server responds with **SYN-ACK** (Synchronize + Acknowledgment).
- **Step 3:** Client sends back an **ACK** (Acknowledgment).

This process ensures both sides are ready and agree on initial sequence numbers.

2. TCP Connection Termination — The 4-Way Handshake

To close a connection cleanly, TCP usually performs a **four-way handshake**:

- One side sends a **FIN** (Finish) flag.
- The other side responds with an **ACK**.
- The second side then sends its own **FIN** when ready.
- The first side sends a final **ACK**.

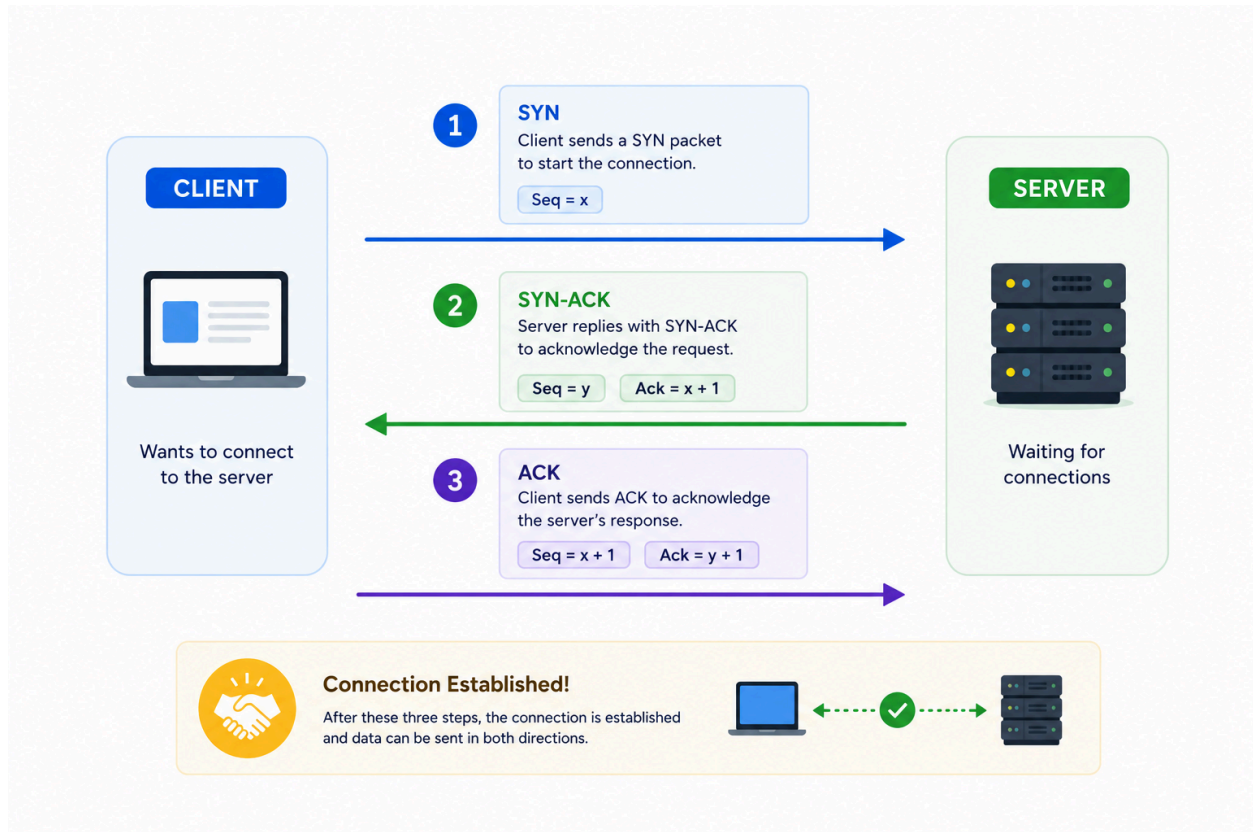
3. Important TCP Fields and Features

- **Ports:** 16-bit source and destination port numbers (multiplexing).
- **Sequence Number:** Tracks the order of bytes sent.
- **Acknowledgment Number:** Indicates the next byte expected.
- **Flags:** Control bits including SYN, ACK, FIN, RST (Reset), PSH (Push), URG (Urgent).
- **Window Size:** Used for flow control (how much data the receiver can accept).

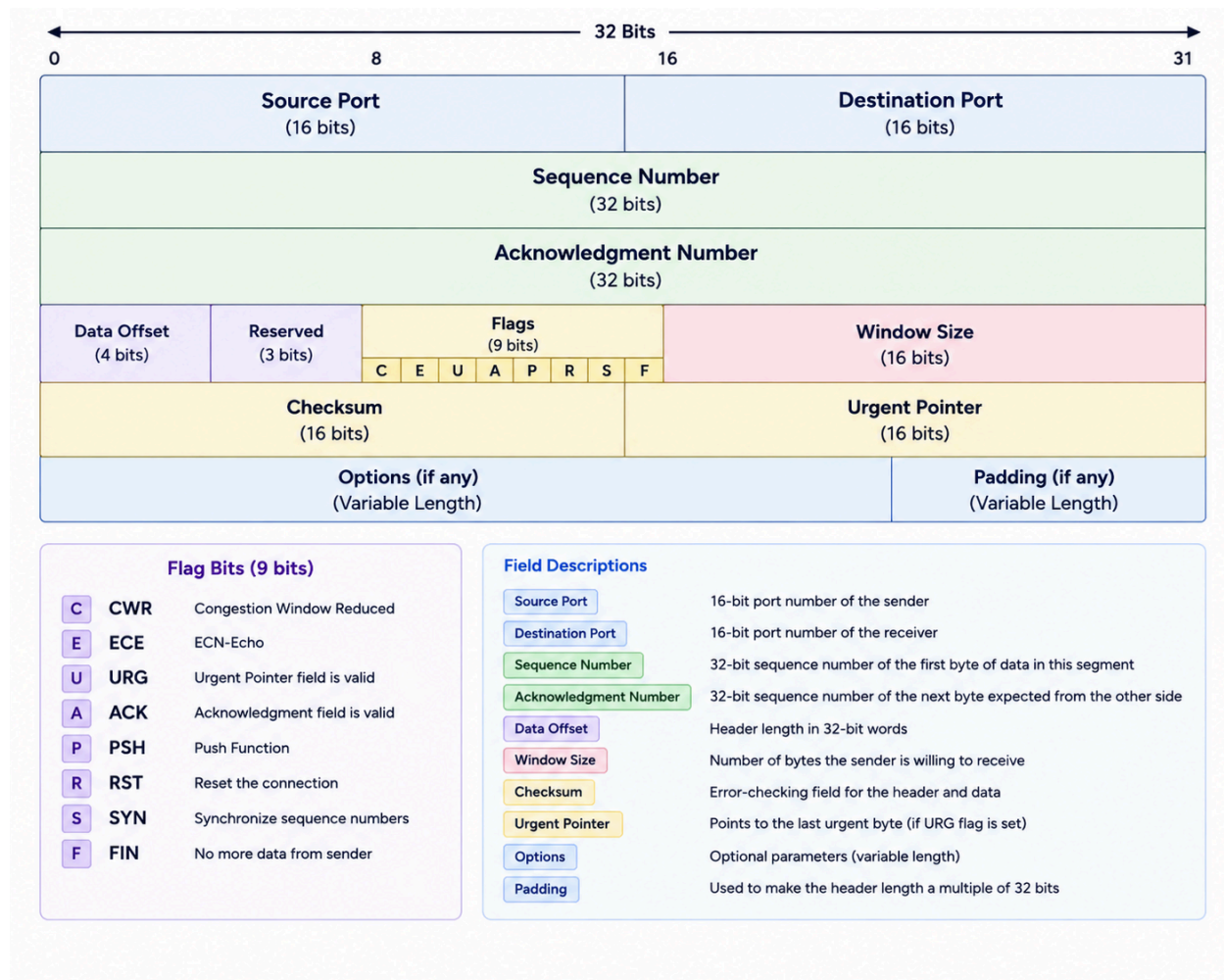
- **Checksum:** Error detection.

Visualizations

TCP 3-Way Handshake:



TCP Header Structure:



MASTER the TCP/IP Model in 5 Minutes!

<https://youtu.be/tK61YFdO3Kw?si=H4bSvzGH0nE5DzuX>